Mapping and Predicting Depression: Bridging Psychometric Networks with Explicit Traits and Implicit Behaviors

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Abstract

This study investigates the structural and behavioral dynamics of depression by integrating Psychometric Network Analysis (PNA) with predictive machine learning, framed within Dual-Process Theory. Explicit symptom evaluations (System 2) were analyzed alongside implicit reaction times (System 1). Structural modeling identified "hopelessness" as the central hub of the depressive network, with stress symptoms occupying an intermediary position between depression and anxiety. Predictively, an XGBoost ablation study demonstrated that incorporating implicit reaction times improves depression severity predictions by over 22.4% compared to standard demographic and personality baselines. Furthermore, SHAP analysis revealed that faster response latencies to core affective symptoms were associated with higher distress. Ultimately, these findings confirm that behavioral latencies capture additional information associated with psychological severity beyond traditional explicit questionnaires.

1 Introduction

This project investigates the structural and behavioral dynamics of depression by combining explicit self-evaluations with implicit behavioral data. This is framed within two models: the Psychometric Network Approach (PNA) to map symptom interactions [1], and Dual-Process Theory to differentiate controlled cognitive reflection (System 2) from automatic reactions (System 1) [5].

By bridging structural mapping with predictive modeling, this project investigates how distress symptoms interconnect, and evaluates whether implicit reaction times can predict depression severity beyond standard demographic profiles.

2 Data Selection

To answer these questions, this project utilizes a public dataset from Open Psychometrics containing responses to the Depression Anxiety Stress Scales (DASS-42) [7], collected between 2017 and 2019. Crucially, the dataset is restricted to participants who explicitly consented to research use and confirmed the accuracy of their answers through the final consent question on research use.

The original dataset contains 39,775 observations. To ensure data quality, observations were removed if participants failed vocabulary validity checks, completed the survey in unrealistic times (< 60s), or provided illogical demographic values (age> 99 and family size> 13). Item-level reaction time outliers (< 0.5s or > 5m) were dropped, and all latencies were subsequently log-transformed to reduce skewness (see **Figure 1**).

After processing, the final dataset consists of 22,848 valid observations. The sample is predominantly young, with a mean age of 23.31 and a median age of 21.0, and culturally diverse, with a notable prevalence of Asian demographics (60.76%) and Muslim religion (58.02%). Gender distribution is predominantly female (77.3%), with the presence of a very small subset identifying as a gender minority, "Other" (1.3%). Exploratory analysis revealed that while male and female baseline depression scores were comparable, at 20.1 and 21.1 respectively, the gender minority group exhibited a higher average of 27.0. Socioeconomically, an inverse relationship emerged between educational attainment and psychological distress: individuals with less than a high school education reported the highest depression scores (24.2), while graduates reported the lowest (18.0).

To operationalize the dual-process framework, variables were partitioned into three subsets: explicit self-reported answers (0–3 Likert scale, denoted by suffix 'A') to model the structural network; implicit reaction times (milliseconds per question, suffix 'E') to serve as behavioral features; and demographic/personality traits (including TIPI scores [6]) to establish foundational baselines for predictive analysis.

3 Analysis Plan

The analysis was conducted in two primary stages, integrating explanatory network analysis with predictive machine learning.

3.1 Step 1: Exploratory Network Modeling (Psychometric Network Approach)

To map symptom interactions, the 42 explicit answers (Q_A) were analyzed. A sparse network was generated using the graphical Lasso algorithm with 5-fold cross-validation [4]. To ensure interpretability, edges with a partial correlation magnitude below 0.05 were thresholded. Finally, degree centrality was calculated to identify the most deeply embedded symptom.

3.2 Step 2: Predictive Modeling and Explainable AI (XAI)

An XGBoost Regressor was trained to predict the continuous total Depression Subscale Score (0-42) [7, 2], calculated from the items listed in **Table 1**. Utilizing a 70/15/15 split for training, validation, and testing, an ablation study was conducted to compare three models:

- **Model A (System 2 baseline):** Trained exclusively on stable user traits, including demographics and TIPI personality scores.
- **Model B (System 1 only):** Trained exclusively on the implicit reaction times (Q_E).
- **Model C (hybrid full model):** Trained on demographics, personality, and reaction times combined.

Models were independently tuned using GridSearch with cross-validation and early stopping. SHAP (SHapley Additive exPlanations) summary and Partial Dependence Plots (PDPs) [8] were subsequently

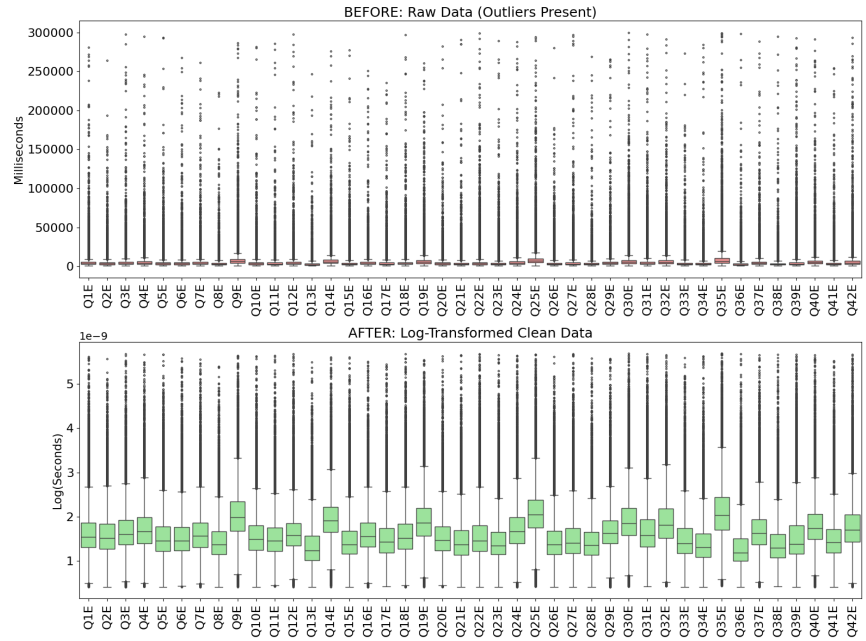

Figure 1: Effect of log-transforming reaction times to reduce latency skewness.

used to interpret the hybrid model and uncover nonlinear relationships.

4 Results

4.1 Structural Results (PNA)

The graphical Lasso algorithm generated a sparse network that naturally clustered into the three distinct subscales: Depression, Anxiety, and Stress. Topologically, the Stress cluster occupies an intermediary position between Depression and Anxiety. Notably, there are no direct edges connecting the Depression and Anxiety clusters; instead, they are connected through the Stress cluster (**Figure 2**). Q8A serves as a bridge between Depression and Stress, while Q33A anchors the Stress–Anxiety boundary. This topological pattern aligns with the item allocation: Q33A belongs to the Stress subscale. Finally, the analysis identified Q10A ("I felt that I had nothing to look forward to"), which aligns with the clinical construct of hopelessness as defined in the DASS manual [7], as the node with the highest degree centrality, indicating that it is the most central symptom within the network.

4.2 Predictive Results (Ablation Study)

The XGBoost models demonstrated varying levels of predictive power for the depression score, measured on a clinical scale from 0 to 42. Mean Absolute Error (MAE) was used to represent the average difference between the predicted and actual scores (**Figure 3**).[1]

- **Model A (System 2: demographics + personality)** achieved an R^2 of 0.3867 (95% CI: 0.3633–0.4087) and an MAE of 7.78 (95% CI: 7.63–7.95).
- **Model B (System 1: reaction times only)** showed significant predictive utility with an R^2 of 0.2238 (95% CI: 0.2017–0.2460) and an MAE of 8.88 (95% CI: 8.71–9.05), indicating that behavioral latencies alone explain a meaningful proportion (20%) of the variance in depression severity.

[1] All confidence intervals were estimated using 10,000 bootstrap iterations.

Table 1: Classification of DASS-42 items by subscale.

Subscale	Questions
Depression	Q3, Q5, Q10, Q13, Q16, Q17, Q21, Q24, Q26, Q31, Q34, Q37, Q38, Q42
Anxiety	Q2, Q4, Q7, Q9, Q15, Q19, Q20, Q23, Q25, Q28, Q30, Q36, Q40, Q41
Stress	Q1, Q6, Q8, Q11, Q12, Q14, Q18, Q22, Q27, Q29, Q32, Q33, Q35, Q39

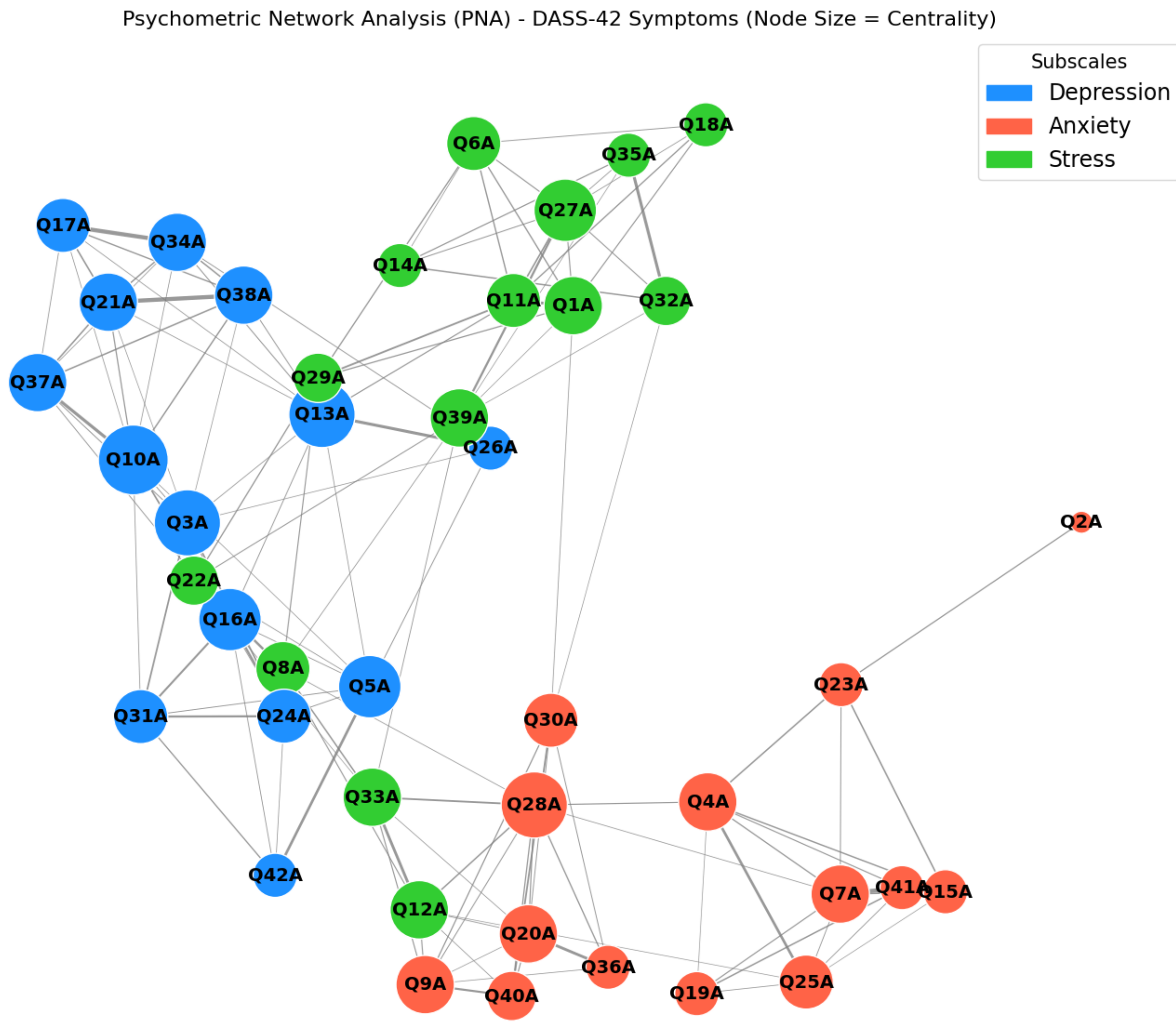

Figure 2: Psychometric network structure estimated from explicit DASS-42 answers.

- **Model C (Hybrid: full model)** achieved the highest performance, with an R^2 of 0.4734 (95% CI: 0.4505–0.4942) and an MAE of 7.06 (95% CI: 6.91–7.21).

Integrating reaction times increased the proportion of explained variance by 8.7 percentage points, a relative improvement of 22.4%, compared to the demographic and personality baseline.

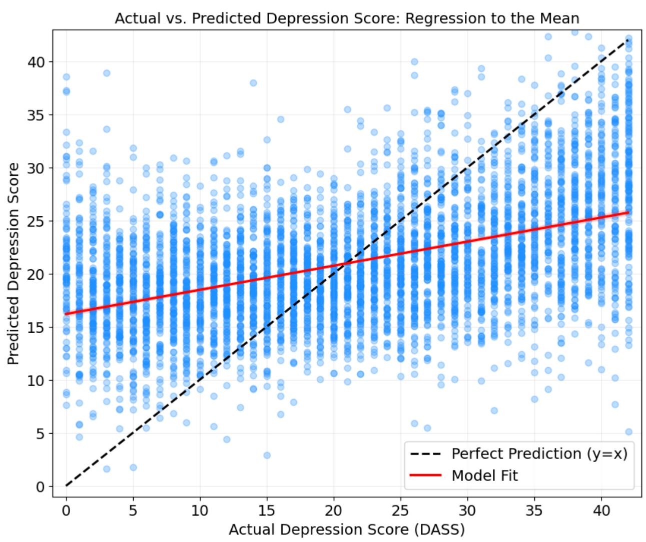

Figure 3: Actual versus predicted depression scores for the hybrid XGBoost model.

4.3 Explainable AI Results (SHAP)

The SHAP summary plot for the hybrid model (**Figure 4**; see Tables 2 and 3 in **Appendix A**) highlighted specific personality traits (TIPI scores) among the strongest baseline predictors, closely followed by key reaction times, notably Q13E and Q23E. Q13 represents a direct recognition of dysphoria, "I felt sad and depressed," while Q23 captures autonomic arousal, "I had difficulty in swallowing." Further down the top features, Q10E, the reaction time to hopelessness, emerged as a key feature, displaying a distinct non-linear impact on the model's predictions.

Beyond reaction times, SHAP analysis revealed distinct sociodemographic gradients (**Figure 5**). Age showed a non-linear impact, elevating predictions for younger individuals and acting as a protective factor for older groups. Lower educational attainment and urban residence consistently pushed predictions higher, whereas graduate-level education served as a strong protective factor. SHAP analysis revealed a distinct gender gradient: female identity acted as a slight protective factor (yielding negative SHAP values), whereas male identity consistently pushed predicted distress higher. Notably, identifying as a gender minority ('Other') produced the highest positive SHAP values, confirming a strong, independent predictive effect for this group. Finally, personality traits heavily influenced predictions. Anxious traits (TIPI4) strongly drove higher distress scores, whereas emotional stability (TIPI9) served as a primary protective factor.

5 Interpretation

Firstly, the structural analysis highlights the utility of the PNA in mapping psychological comorbidity [3]. The network's topology demonstrates how psychological domains overlap: rather than being entirely distinct entities, Anxiety and Depression are structurally bridged by Stress. This aligns with the PNA hypothesis that comorbidity arises from direct symptom-to-symptom interactions, with specific stress responses serving as pathways connecting depressive and anxiety symptoms.

Complementing the structural insights, the predictive phase is consistent with the Dual-Process framework in mental health modeling. The improvement in predictive accuracy suggests that response latencies capture additional variance in distress not explained by demographic and personality variables.

Furthermore, the study revealed a dynamic between explicit structure and implicit behavior. In the explicit network, Q10A (hopelessness) emerged as the central hub within the depressive network. However, the strongest automatic behavioral predictors in the XGBoost model were reaction times to core affective and somatic symptoms, specifically Q13E

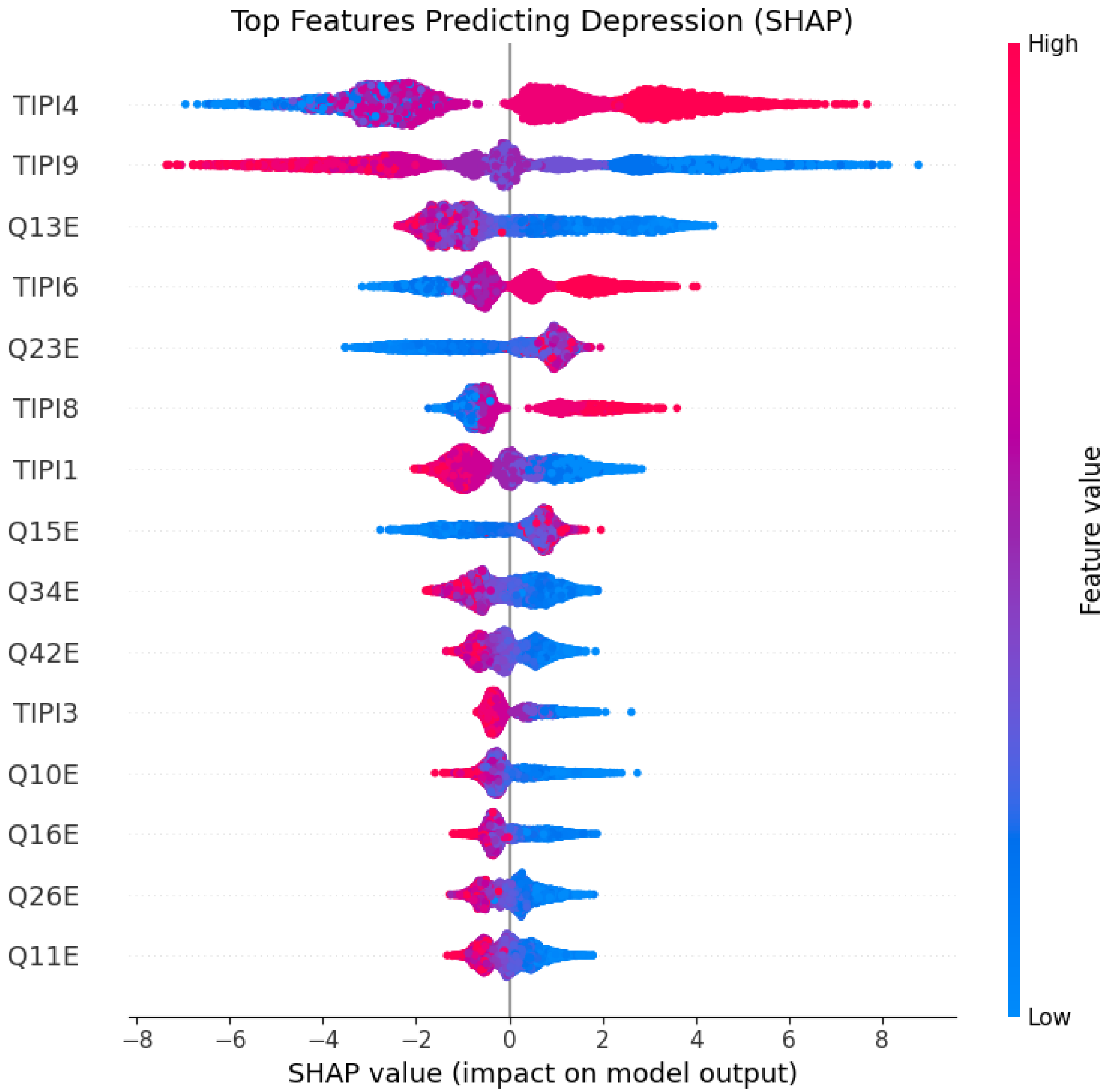

Figure 4: SHAP summary plot for the hybrid model, showing the strongest personality and reaction-time predictors.

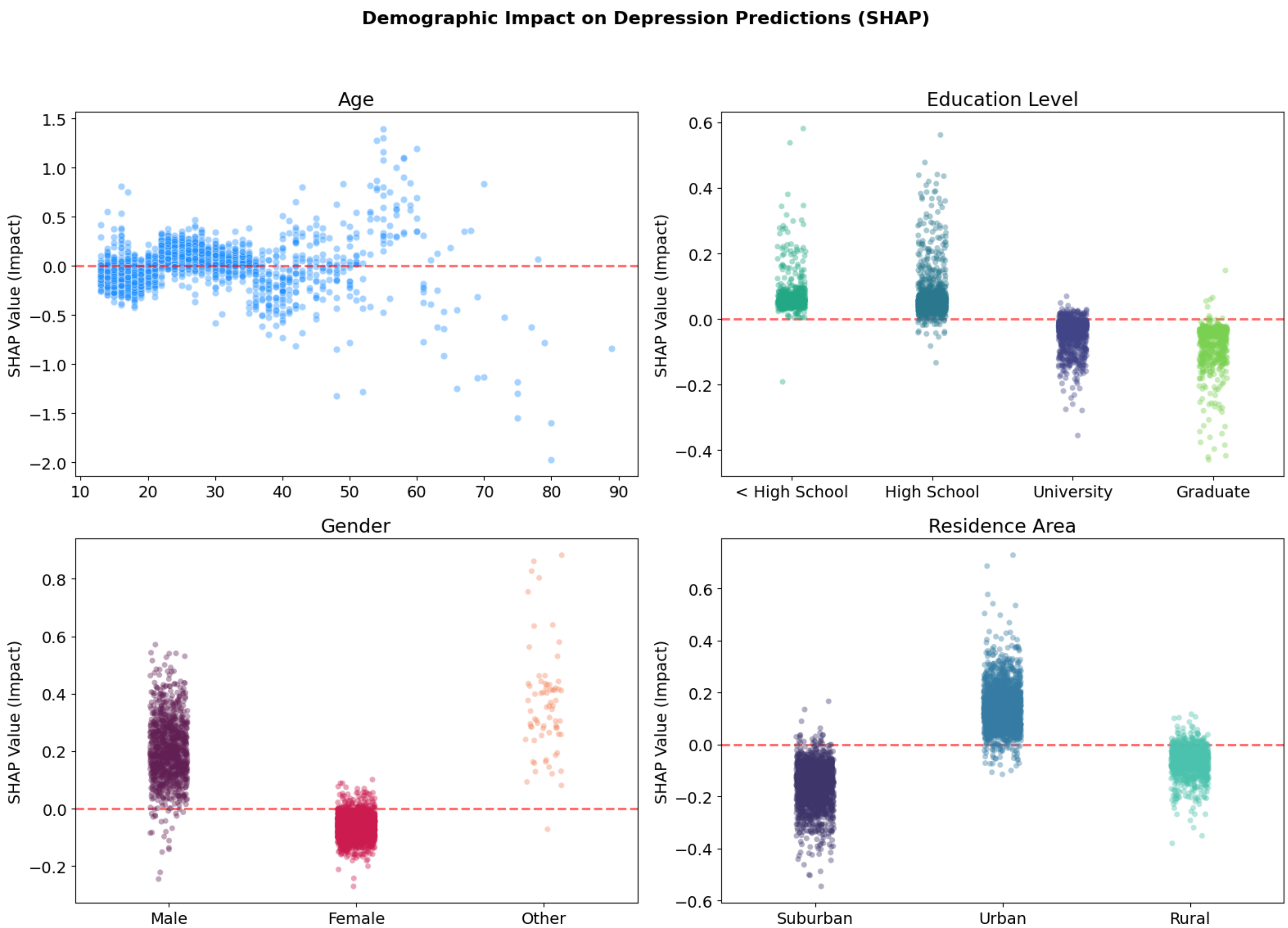

Figure 5: Sociodemographic SHAP effects for age, education, gender, and residence area in the hybrid model.

and Q23E. This suggests that while hopelessness provides the structural foundation of the depressive network, the cognitive accessibility of sadness and physical anxiety provides the strongest behavioral predictors. Yet, the reaction time to hopelessness (Q10E) still provided crucial behavioral insights. The SHAP dependence plot for Q10E (**Figure 6**) reveals a distinct non-linear pattern: shorter response latencies for hopelessness were associated with higher predicted depression severity. Conversely, longer latencies were associated with lower predicted depression scores and reduced distress.

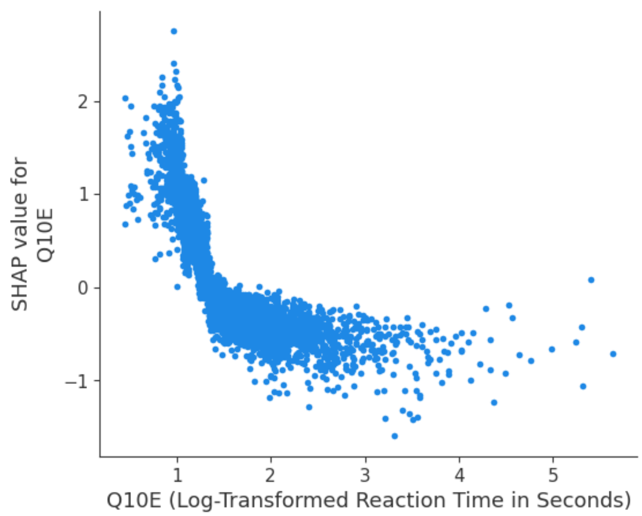

Figure 6: Non-linear effect of Q10E, the reaction time to hopelessness, on predicted depression severity.

Beyond individual behaviors, the SHAP analysis confirms that systemic socioeconomic disadvantages independently shape a baseline of distress. Notably, the model's behavior aligns with Minority Stress Theory [9, 10], as identifying as a gender minority significantly elevated predicted distress. Interestingly, although females showed slightly higher raw depression scores than males (21.1 vs. 20.1), SHAP reversed this marginal pattern after adjustment, suggesting that the female raw-score difference may be explained by correlated traits or reaction-time features. Crucially, the model utilizes these stable traits to establish a foundational sociodemographic risk profile, which is then dynamically modulated by the implicit reaction-time features.

6 Limitations

The dataset exhibits a strong demographic skew, being predominantly young, female, and concentrated within Asian and Muslim populations, which may limit generalizability to other demographic groups.

Reaction times also introduce substantial noise, as this web-based dataset cannot control for factors such as device type, distractions, or fatigue that affect response speed independently of psychological distress.

Additionally, because reaction times are derived from the same questionnaire items used to calculate depression scores, part of the model's predictive power may reflect item-level behavioral coupling rather than independent behavioral markers. Future work could assess this by excluding Depression item latencies.

Finally, the model demonstrates a regression-to-the-mean effect, overestimating low depression scores and underestimating high ones. This suggests that while demographic and reaction-time data capture general distress patterns, they are insufficient for accurately predicting clinical severity, which is likely influenced by unmeasured factors beyond the survey's scope.

Appendix A: Feature Dictionary

Note: In the dataset, DASS-42 questions are denoted with the suffix 'A' for the explicit answer (e.g., Q10A) and 'E' for the implicit reaction time (e.g., Q10E).

Code	DASS Scale	Original Text	Interpretation
Q8	Stress	"I felt it was difficult to relax"	Difficulty Relaxing
Q10	Depression	"I felt that I had nothing to look forward to"	Hopelessness
Q11	Stress	"I found myself getting agitated"	Agitation
Q13	Depression	"I felt sad and depressed"	Dysphoria
Q15	Anxiety	"I felt I was close to panic"	Panic-like Anxiety
Q16	Depression	"I was unable to become enthusiastic about anything"	Lack of Interest/Involvement
Q23	Anxiety	"I had difficulty in swallowing"	Autonomic Arousal
Q26	Depression	"I felt down-hearted and blue"	Dysphoria
Q33	Stress	"I was in a state of nervous tension"	Nervous Tension
Q34	Depression	"I felt I was pretty worthless"	Self-deprecation
Q42	Depression	"I found it difficult to work up the initiative to do things"	Lack of Initiative

Table 2: DASS-42 question dictionary for features highlighted in the structural network (PNA) and predictive (SHAP) analyses. The "interpretation" label reflects the symptom clusters described in the DASS-42 manual [7].

Item	Big Five Trait	Original Text	Score direction
TIPI 1	Extraversion	"Extraverted, enthusiastic"	Positive
TIPI 6	Extraversion	"Reserved, quiet"	Reverse-coded
TIPI 3	Conscientiousness	"Dependable, self-disciplined"	Positive
TIPI 8	Conscientiousness	"Disorganized, careless"	Reverse-coded
TIPI 4	Emotional Stability	"Anxious, easily upset"	Reverse-coded
TIPI 9	Emotional Stability	"Calm, emotionally stable"	Positive

Table 3: TIPI personality trait dictionary for features discussed in the analysis.

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